

① Find $\frac{df}{dt}$ for the following functions f :

- directly
- chain rule

Seminar 10 (H.A.)

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a) $f(x,y) = \ln(x^2 + y^2)$

$x(t) = t$

$y(t) = t^2$

$t \mapsto (x(t), y(t)) \xrightarrow{f} f(x,y)$

$\frac{u'}{u}$

Directly: $f(t) = \ln(t^2 + t^4)$

$f'(t) = \frac{df}{dt} = \frac{(t^2 + t^4)'}{t^2 + t^4} = \frac{2t + 4t^3}{t^2 + t^4}$

Chain rule: 1d: $f(u) \rightarrow f(u(t))$, $\frac{df}{dt} = \frac{df}{du} \cdot \frac{du}{dt} = \dots$ $t \rightarrow u \rightarrow f(u)$
 $t \rightarrow u(t) \rightarrow f(u(t))$

$(f(u(t)))' = f'(u) \cdot u'$

Chain rule: 2d: $\boxed{\frac{df}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt}}$

$\frac{\partial f}{\partial x} = \frac{2x}{x^2 + y^2}$ $\rightarrow$ partial derivative (depends on $>$ variables)

$\frac{\partial f}{\partial y} = \frac{2y}{x^2 + y^2}$

$\frac{dx}{dt} = 1$ ($x(t) = t$)

$\frac{dy}{dt} = 2t$ ($y(t) = t^2$)

$\Rightarrow \frac{df}{dt} = \frac{2x}{x^2 + y^2} \cdot 1 + \frac{2y}{x^2 + y^2} \cdot 2t = \frac{2t}{t^2 + t^4} + \frac{4t^3}{t^2 + t^4} = \frac{2t + 4t^3}{t^2 + t^4}$

$\downarrow$
I'm looking at f as a function of $t \rightarrow \underline{(f(t))'}$ (depends on one variable)

Chain rule 3d:

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial f}{\partial z} \cdot \frac{dz}{dt}$$

b) $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$

$$x(t) = \cos t$$

$$y(t) = \sin t$$

$$z(t) = t > 0.$$

directly

$$t \mapsto (x, y, z) \mapsto \underline{f(x, y, z)}.$$

Directly: remove $x, y, z \rightarrow$ rewrite everything only using t .

$$f(t) = \sqrt{\cos^2 t + \sin^2 t + t^2} = \sqrt{1+t^2}$$

$$\frac{df}{dt} = (\sqrt{1+t^2})' = \frac{(1+t^2)'}{2\sqrt{1+t^2}} = \frac{2t}{2\sqrt{1+t^2}} = \frac{t}{\sqrt{1+t^2}} = \underline{f'(t)}.$$

Chain rule: $\frac{\partial f}{\partial x} = \frac{2x}{2\sqrt{x^2+y^2+z^2}} = \frac{x}{\sqrt{x^2+y^2+z^2}}$

$\frac{d}{dt}$ ct.

$$\frac{\partial f}{\partial y} = \frac{y}{\sqrt{x^2+y^2+z^2}}$$

$$\frac{\partial f}{\partial z} = \frac{z}{\sqrt{x^2+y^2+z^2}}$$

$$\frac{dx}{dt} = (x(t))' = (\cos t)' = -\sin t.$$

$$\frac{dy}{dt} = (y(t))' = (\sin t)' = \cos t$$

$$\frac{dz}{dt} = (z(t))' = t' = 1.$$

$$\Rightarrow \frac{df}{dt} = \frac{x}{\sqrt{x^2+y^2+z^2}} \cdot (-\sin t) + \frac{y}{\sqrt{x^2+y^2+z^2}} \cdot \cos t + \frac{z}{\sqrt{x^2+y^2+z^2}}$$
$$= \frac{-t \sin t + t \cos t + t}{\sqrt{x^2+y^2+z^2}}$$

?

$$= \frac{-\cos t \cdot \sin t}{\sqrt{1+t^2}} + \frac{\sin t \cdot \cos t}{\sqrt{1+t^2}} + \frac{t}{\sqrt{1+t^2}} = \frac{t}{\sqrt{1+t^2}}$$

② Let $f(u, r, w) = u^2 + r^2 - w$
 $u(x, y, z) = x^2 y$
 $r(x, y, z) = y^2$
 $w(x, y, z) = e^{-xz}$

Chain Rule
 $(x, y, z) \mapsto (u, r, w) \xrightarrow{f} f(u, r, w)$

Find $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$, $\frac{\partial f}{\partial z}$ in 2 ways.

Directly: $f(x, y, z) = x^4 y^2 + y^4 - e^{-xz}$

$$\frac{\partial f}{\partial x} = 4x^3 y^2 + y^4 + z e^{-xz}$$

$$\frac{\partial f}{\partial y} = 2x^4 y + y^4 - e^{-xz}$$

$$\frac{\partial f}{\partial z} = x^4 y^2 + y^4 + x \cdot e^{-xz}$$

Chain rule:

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial f}{\partial w} \cdot \frac{\partial w}{\partial x}$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial f}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial f}{\partial w} \cdot \frac{\partial w}{\partial y}$$

$$\frac{\partial f}{\partial z} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial z} + \frac{\partial f}{\partial r} \cdot \frac{\partial r}{\partial z} + \frac{\partial f}{\partial w} \cdot \frac{\partial w}{\partial z}$$

$$f(u, r, w) = u^2 + r^2 - w$$

$$\begin{cases} \frac{\partial f}{\partial u} = 2u \\ \frac{\partial f}{\partial r} = 2r \\ \frac{\partial f}{\partial w} = -1 \end{cases}$$

$$\begin{cases} u = x^2 y \\ r = y^2 \\ w = e^{-xz} \end{cases}$$

$$\begin{cases} \frac{\partial u}{\partial x} = 2xy \\ \frac{\partial r}{\partial x} = 0 \end{cases}$$

$$\frac{\partial w}{\partial x} = -2 \cdot e^{-xz}$$

$$\begin{aligned} \frac{\partial f}{\partial x} &= 2u \cdot 2xy + 2r \cdot 0 + z \cdot e^{-xz} \\ &= 4x^2 y \cdot xy + 0 + z \cdot e^{-xz} \\ &= 4x^3 y^2 + z e^{-xz} \quad (y, z \text{ ct.}) \end{aligned}$$

Def: Directional derivatives $Df_r(x) = \lim_{h \rightarrow 0} \frac{f(x+hr) - f(x)}{h} = \nabla f(x) \cdot r$.
 (derivative in a certain direction) Scalar prod.

③ Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = x^2 + xy$

a) Find the gradient and the direction of steepest descent at $(1, 0)$.

b) Find the derivative at $(1, 0)$ in the direction of $(1, 1) = e_1 + e_2$.

a) $\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = (2x + y, x)$

$\nabla f(1, 0) = (2, 1)$.

Direction of steepest descent: $-\nabla f(1, 0) = (-2, -1)$.

b) $D_{\frac{(1,1)}{\sqrt{2}}}(1, 0) = \nabla f(1, 0) \cdot \frac{(1, 1)}{\sqrt{2}} = (2, 1) \cdot \frac{(1, 1)}{\sqrt{2}} = \frac{2+1}{\sqrt{2}} = \frac{3}{\sqrt{2}}$.

④ Let $f: \mathbb{R}^m \rightarrow \mathbb{R}$, $f(x) = \frac{1}{2} \|x\|^2$.

a) Find $\nabla f(x)$ (gradient of f at a point x)

b) Find $Df_r(x)$, $r \in \mathbb{R}^m$, in 2 ways.

a) $\|x\|^2 = \langle x, x \rangle = x_1^2 + x_2^2 + \dots + x_m^2$

$f(x) = \frac{1}{2} (x_1^2 + x_2^2 + \dots + x_m^2)$

$\nabla f = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_m} \right) = (x_1, x_2, \dots, x_m) = x$

$\frac{\partial f}{\partial x_1} = \frac{1}{2} \cdot x_1^2 + \frac{1}{2} x_2^2 + \dots = \frac{1}{2} \cdot 2 \cdot x_1 = x_1$.

$\frac{\partial f}{\partial x_m} = x_m$.

$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}(x), \frac{\partial f}{\partial x_2}(x), \dots, \frac{\partial f}{\partial x_m}(x) \right) = (x_1, \dots, x_m) = x$.

b) $Df_r(x) = \nabla f(x) \cdot r = x \cdot r = \langle x, r \rangle$

scalar prod of 2 vectors.

mor?, mor?

"derivative of f in the direction of r "

With the definition: (lim)

$$Df_r(x) = \lim_{h \rightarrow 0} \frac{f(x+hr) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{2} \|x+hr\|^2 - \frac{1}{2} \|x\|^2}{h} =$$

$$= \frac{1}{2} \lim_{h \rightarrow 0} \frac{\|x+hr\|^2 - \|x\|^2}{h} = \frac{1}{2} \lim_{h \rightarrow 0} \frac{\langle x+hr, x+hr \rangle - \langle x, x \rangle}{h} =$$

$$= \frac{1}{2} \lim_{h \rightarrow 0} \frac{\langle x, x \rangle + h \langle x, r \rangle + h \langle r, x \rangle + h^2 \langle r, r \rangle - \langle x, x \rangle}{h}$$

$$= \frac{1}{2} \lim_{h \rightarrow 0} \frac{2h \langle x, r \rangle + h^2 \langle r, r \rangle}{h}$$

$$= \frac{1}{2} \lim_{h \rightarrow 0} \frac{2 \langle x, r \rangle + h \langle r, r \rangle}{1}$$

$$= \langle x, r \rangle + \frac{1}{2} \lim_{h \rightarrow 0} h \langle r, r \rangle$$

$$= \langle x, r \rangle + 0$$

$$= \langle x, r \rangle$$

2nd order derivatives:

differentiate with respect to x_i first!

closer to the function

$$\frac{\partial^2 f}{\partial x_j \partial x_i} = \frac{\partial}{\partial x_j} \left(\frac{\partial f}{\partial x_i} \right) \quad i, j = \overline{1, m}$$

HESSIAN MATRIX

(SYMMETRIC)

$$H(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_1 \partial x_3} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_m} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2 \partial x_2} & \frac{\partial^2 f}{\partial x_2 \partial x_3} & \dots & \frac{\partial^2 f}{\partial x_2 \partial x_m} \\ \frac{\partial^2 f}{\partial x_3 \partial x_1} & \dots & \dots & \dots & \frac{\partial^2 f}{\partial x_3 \partial x_m} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_m \partial x_1} & \dots & \dots & \dots & \frac{\partial^2 f}{\partial x_m \partial x_m} \end{bmatrix}$$

5. Let $D = \text{diag}(d_1, \dots, d_m) = \begin{bmatrix} d_1 & & 0 \\ 0 & d_2 & \\ \vdots & & \ddots \\ 0 & & & d_m \end{bmatrix}$

$$f: \mathbb{R}^m \rightarrow \mathbb{R}, f(x) = \frac{1}{2} x^T D x$$

transposed vector (column $\rightarrow$ row
row $\rightarrow$ column)

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}_{m \times 1}$$

$m \times 1$ $m \times 1$

$$x^T = [x_1 \dots x_m]_{1 \times m}$$

$$f(x) = \frac{1}{2} [x_1 \dots x_m] \cdot \underbrace{\begin{bmatrix} d_1 & & & \\ & d_2 & & \\ & & \ddots & \\ 0 & & & d_m \end{bmatrix}}_{= \begin{bmatrix} d_1 x_1 \\ \vdots \\ d_m x_m \end{bmatrix}} \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} = \frac{1}{2} [x_1 \dots x_m] \cdot \begin{bmatrix} d_1 x_1 \\ \vdots \\ d_m x_m \end{bmatrix} = \frac{1}{2} (x_1^2 \cdot d_1 + x_2^2 \cdot d_2 + \dots + x_m^2 \cdot d_m)$$

$$\nabla f(x) = ?$$

$$\frac{\partial f}{\partial x_1} = d_1 \cdot x_1$$

$$\frac{\partial f}{\partial x_i} = d_i \cdot x_i$$

$$\Rightarrow \nabla f(x) = (d_1 x_1, \dots, d_m x_m) = \underline{D \cdot x} \quad \text{linear function}$$

(The derivative is a vector)

$\begin{cases} \text{vector} \rightarrow 1 \text{ index} \\ \text{matrix} \rightarrow 2 \text{ indices} \end{cases} \rightarrow \text{TENSORS (machine learning)}$

$$H(x) = ?$$

$$\frac{\partial^2 f}{\partial x_i \partial x_i} = d_i$$

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = 0, j \neq i. \quad \frac{\partial^2 f}{\partial x_1 \partial x_2} = 0$$

$$\frac{\partial^2 f}{\partial x_i \partial x_j}$$

$$H(x) = \begin{bmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & & \ddots & \\ 0 & \dots & & d_m \end{bmatrix} = \underline{D} \quad \text{constant matrix.}$$